DSA Assignment

Experience:

I found the data structures and algorithms course to be actually quite fun but also a bit tiring and boring at times since it was mostly always invisible and hard to visualise it myself. The theory itself behind everything was actually quite easy to get a hang on after reading through it twice or thrice.

Challenges:

The challenges I faced the most were actually most of the time just small details I missed when writing the code, like for example the “=” sign that I forgot about 6 times during different comparisons. My solution to this problem was to repeatedly mindlessly try to solve it for a couple hours, sprinkled with some procrastination before either doing it manually line by line and on paper then finding the problem or asking my teacher Jo for help where we finally found it after a bit.

The only other main challenges I have faced is drifting off and writing my lefts and rights wrong (Week 5s work on BFS) but also failing to do simple math (Week 4s work) as I on some days was sleep deprived from watching Cyberpunk: Edgerunners or from getting a random rush of energy to be productive which led me to sleep extremely late. The solution to this problem was coming in the day after with a brain of either full sleep or waking up on the correct side of the bed and instantly noticing the problems at hand. And when that didn’t work I would usually turn to my teacher and explain my brain is not currently functioning at proper capacity.

Decisions:

For week 4 I decided to make the obvious decision of picking Binary search which was the lowest complexity out of the bunch. It was also the easiest to understand since I was already familiar with it from cheating on guessing games with my family by using it.

Then as for week 5 I decided to implement BST which made the most sense to me and was the easiest to visualise. Additionally I also removed the insert function since my binary search tree was made in a recursive function instead where I fed it the whole array. That said it is still possible to add in the future if necessary as it is compatible.

The way week 5's function that makes the tree works is by taking in a sorted list then recursively selecting the median as the left and right branches. Since it is sorted already all I had to do is take the left sub-medians and put them to the left branches and the right sub-medians and put them with the right branches. I made this function based on the binary search from the week before. This structure also allows the use of binary search coming in through the root by taking a left if its smaller and right if its larger.

Results:

The suspects I narrowed it down to are:

- Nem Wellington
- Helena Blackwood
- Alfred Pugsley
- Jasper Wellington

My guess here is that Alfred Pugsley is either the victim or the murderer in this scenario as he's the odd one out with having an exit time that is after the enter time, unlike the others.

As for which one is faster between Breadth First Searching in a binary tree and Binary Searching in a sorted array, Binary Searching in a sorted array is ten times faster.